

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester M.Tech (MMD) (CBCS) Degree Examinations
 December 2024 – January 2025
22MMD121 – CONTINUUM MECHANICS

Duration: 3 Hours

Max. Marks: 100

**Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each
 and One full question from Unit – III.**

Unit – I

	Marks	BT*	CO*	PO*
1. a) Arrive at the Cauchy's stress formula equations to determine the stress components on any arbitrary plane.	10	L*1	1	1
b) For the given displacement field $u = c(x^2 + 2x)$, $v = c(4x + 2y^2 + z)$, $w = 4cz^2$ where c is a very small constant, determine the strain at (2,1,3) in a plane having direction cosines $l = 0$, $m = -0.7071$ and $n = 0.7071$.	10	L3	2	2
2. a) Arrive at the strain compatibility equations in 3D.	10	L1	2	1
b) Determine the principal stresses and the plane corresponding to the maximum principal stress for the following state of stress $[\sigma_{ij}]$ at a point.				
$\begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 4 \\ 0 & 4 & -3 \end{pmatrix}$	10	L3	1	2
3. a) The stress field within an elastic structural member is expressed as follows: $\sigma_x = 7x^3 + 2y^2$; $\tau_{xy} = 5z + 3y^2$; $\tau_{xz} = 3xz^3 + x^2y$; $\sigma_y = 5x^3 - 3.5y^2$; $\tau_{yz} = 0$; $\sigma_z = 3y^2 + 2z^3$. Determine whether the body is in equilibrium or not at the point (-1, 5, 3). If not, what should be the body force components which will bring this body into equilibrium?	10	L3	1	2
b) The strain components at a point with respect to x y z co-ordinate system are given by: $\epsilon_x = 0.10$, $\epsilon_y = 0.20$, $\epsilon_z = 0.30$, $\gamma_{xy} = \gamma_{yz} = \gamma_{xz} = 0.160$ If the coordinate axes are rotated about the z-axis through 45° in the anticlockwise direction, determine the new strain components.	10	L3	2	2

Unit – II

4. a) Verify if $\phi = (q / 8c^3) [x^2(y^3 - 3c^2y + 2c^3) - 1/5(y^3(y^2 - 2c^2))]$ is an Airy's stress function or not.	10	L3	3	2
b) Explain (i) Principle of superposition (ii) St. Venant's principle	10	L1	3	1
5. a) Arrive at the Beltrami-Mitchel equation for plane strain condition assuming body forces are present.	10	L1	3	1
b) Arrive at the biharmonic equation for plane stress condition.	10	L1	3	1
6. a) Investigate what problem is solved by the stress function, $\phi = (-F/d^3) xz^2(3d - 2z)$ applied to the region included in $z = 0$ to $z = d$ and $x=0$, on the side x positive.	10	L3	3	2
b) Using the inverse method, obtain the expressions for the stresses in a thin cantilever beam carrying a point load 'P' at its free end. The beam is of rectangular cross-section having width 'b' and depth '2h'. Also, sketch the variations of the stresses at sections away from the ends.	10	L1	3	1

Unit – III

7. a) Develop the strain compatibility equations in polar coordinates.
 b) A steel cylinder which has an inside diameter of 1m is subjected to an internal pressure of 8 MPa. Calculate the wall thickness if the maximum shearing stress is not to exceed 35 MPa.
8. a) A thin walled box section having dimensions $2a \times a \times t$ is to be compared with a solid circular section of diameter 'a' as shown in Fig. Q8a. Determine the thickness 't' so that the two sections have:
 (a) The same maximum shear stress for the same torque.
 (b) The same stiffness.

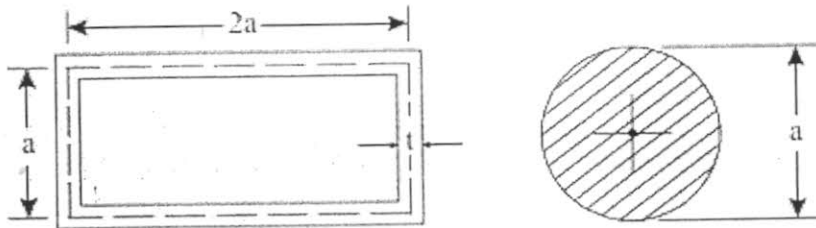


Fig. Q8a

- b) Explain the Maxwell's and Kelvin's models used for modeling visco-elastic behaviour.

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
